

SkyNetix Enterprise Network Design — Cisco Packet Tracer Project

Cleaned Portfolio Documentation

Focus:

- Kuala Lumpur Branch LAN Design**
- Multi-Branch WAN Context**
- VLSM and IP Addressing**
- Cisco Packet Tracer Simulation**
- Network Topology Justification**

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Introduction

SkyNetix Sdn. Bhd. is a prominent enterprise technology company in Malaysia that specializes in networking, IT consulting, and cloud infrastructure, offering digital connectivity and IT support to corporate and GLC companies. The company has a significant national presence with branches in various regions, including Kuala Lumpur, Penang, Johor Bahru, and Kota Kinabalu. These branches serve as representative locations for the company's operations. With over **200 professional** engineers and IT specialists, SkyNetix focuses on combining technical expertise with best practices to help Malaysian enterprises navigate the digital landscape effectively.

As part of its digital expansion, SkyNetix is undertaking a nationwide network infrastructure upgrade across its branches to improve internal communications security and operational efficiency. The company aims to establish a cohesive network that facilitates high-speed data transfers, integrated cloud planning, and backup redundancies for uninterrupted service. The headquarters in **Kuala Lumpur** will oversee operations, while each regional site will undergo infrastructural enhancements to create a unified network. The upgrade is designed to align all branches with common protocols and expectations, utilizing **Cisco Packet Tracer** for network design and simulations.

By utilizing Cisco Packet Tracer, SkyNetix aims to visualize network designs, test performance measures, and develop realistic plans for implementation. This approach not only serves as a technical guide for future upgrades but also as an educational case study on professional development within a corporate setting. Through visual representations and working examples, the paper aims to provide insights into network topology, routing protocols, and security measures, showcasing how companies like SkyNetix maintain security and performance standards across their branches. The modernization of SkyNetix's network infrastructure signifies a crucial step towards technical improvement and operational innovation in the evolving digital landscape.

Kuala Lumpur Branch

Assumptions About The Branch

The Kuala Lumpur City Branch of **SkyNetix Sdn. Bhd.** is designed as a compact, yet high-efficiency corporate office located within Malaysia's capital. Occupying a single-floor unit of approximately **250 m²**, the branch adopts a nearly square layout ideal for interdepartmental collaboration and streamlined workflows.

While not being the largest in SkyNetix's major operational sites, this branch represents a **scaled prototype** for future regional expansions—combining administrative, operational, and technical functions within a digitally integrated workspace.

The office includes:

- **Three** key departments – Administration, Finance, and IT
- A dedicated **Server Room/IT Room**
- A **Meeting Room** Linked to Finance Department
- A **Reception Area**
- A **Resting Area** for employees

The branch operates with **27 personnel**, structured for efficiency and inter-departmental synergy.

The **Administration** Department is the main control center for the entire branch, operating from one location with eight personnel specializing in management and HR. The group consists of the branch manager, operations manager, department director, HR manager, and four HR specialists. Day-to-day evaluations, business trends and confidential meetings occur under one roof while also providing oversight for finance/HR/technical matters as all parties are digitally connected in a safe, branch-wide environment. HR processes (recruitment, onboarding, payroll, and benefits) are connected to the Administration's support with **encrypted file transfer, role-based access, and retention policies** in place for personal data to comply with Malaysia's Personal Data Protection

Act (**PDPA**) for individual transparency and accountability and in accordance with **ISO/IEC 27001** information security management standards for employee data protection (PDPA 2010; ISO 2022).

Finance Department. The Finance Department includes six employees and represents the branch's financial hub—budgetary measures, approval of procurement requests and payroll estimates. SkyNetix HQ houses secured VLANs that connect and encrypt to ensure all appropriate budgetary allocations receive connectivity to the main office for budgetary upkeep. Although most financial endeavors are digitalized, a paper trail exists in secured interdepository rooms for proper handling. SkyNetix intends to go green, however, therefore, older surplus documents are sent to a third-party recycling company to ensure compliance of decommissioned documents which has been known to **decrease business waste-disposal expenses and reduce lifecycle emissions for office paper** (WRAP claims large companies can achieve **£400–£1,000 per employee/year** of savings on office recycling, and the EPA's WARM finds emissions avoided from recycling office paper exceed those from sending it to a landfill).(EPA, 2023)

A glass wall separates the Finance Department from the Reception area, allowing a view of visitors into the dynamic operations of Finance, embodying the company's transparent culture of evolution.

IT Department (Server Room Included). The 12-member IT team covers all technology operations, and the Server Room is next door for immediate troubleshooting and supervision. The team is responsible for **LAN management** and identities; edge security and firewall; database integrity checks; centralized databases for line-of-business applications and backups; branching redundancy (i.e., failover routes, link resiliency) with periodic testing—per campus/branch high availability design standards.(Cisco, 2022)

The IT team works with the IT team in the other branches for baseline images, policies, and constant updates/patches for security compliance and updates across all locations.(Murugiah Souppaya, 2022)

Floor Plan

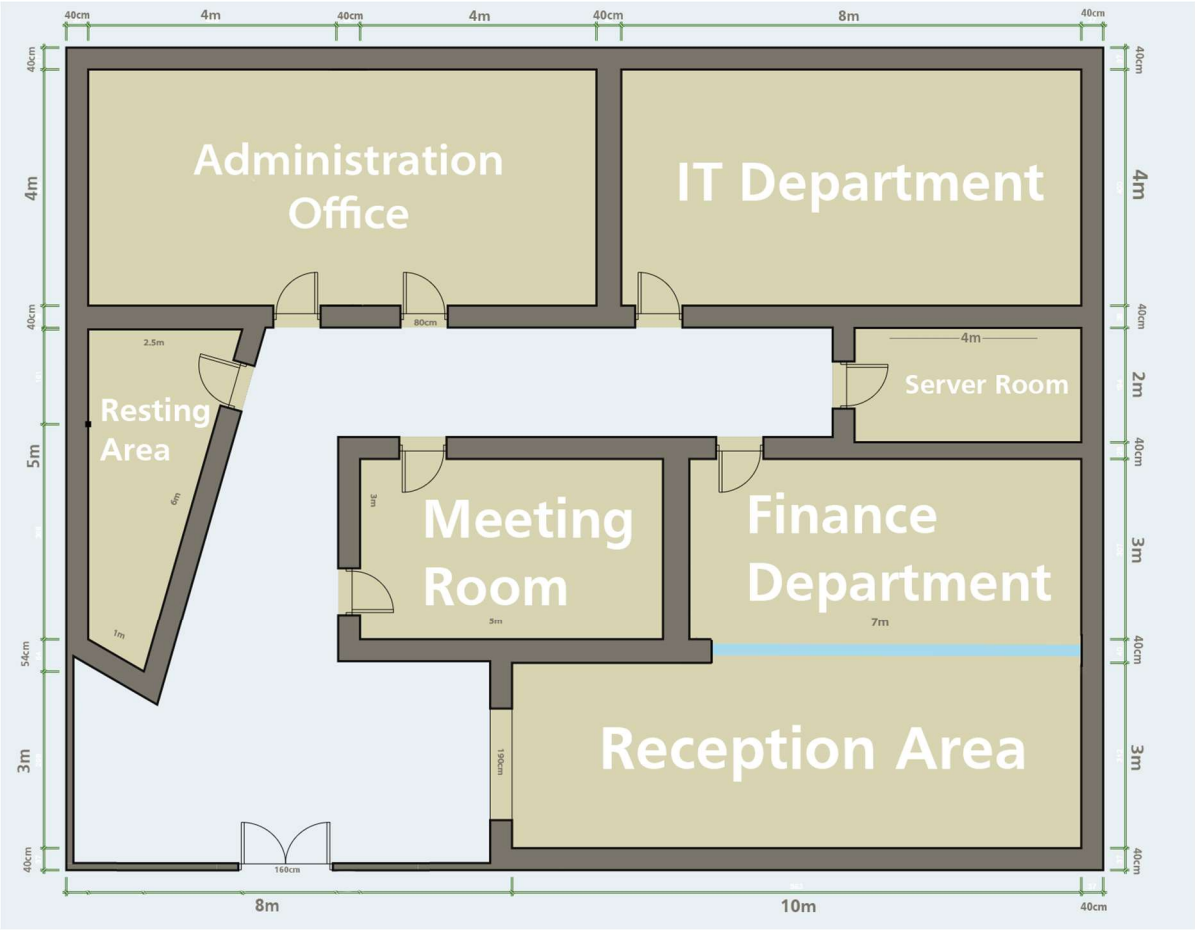
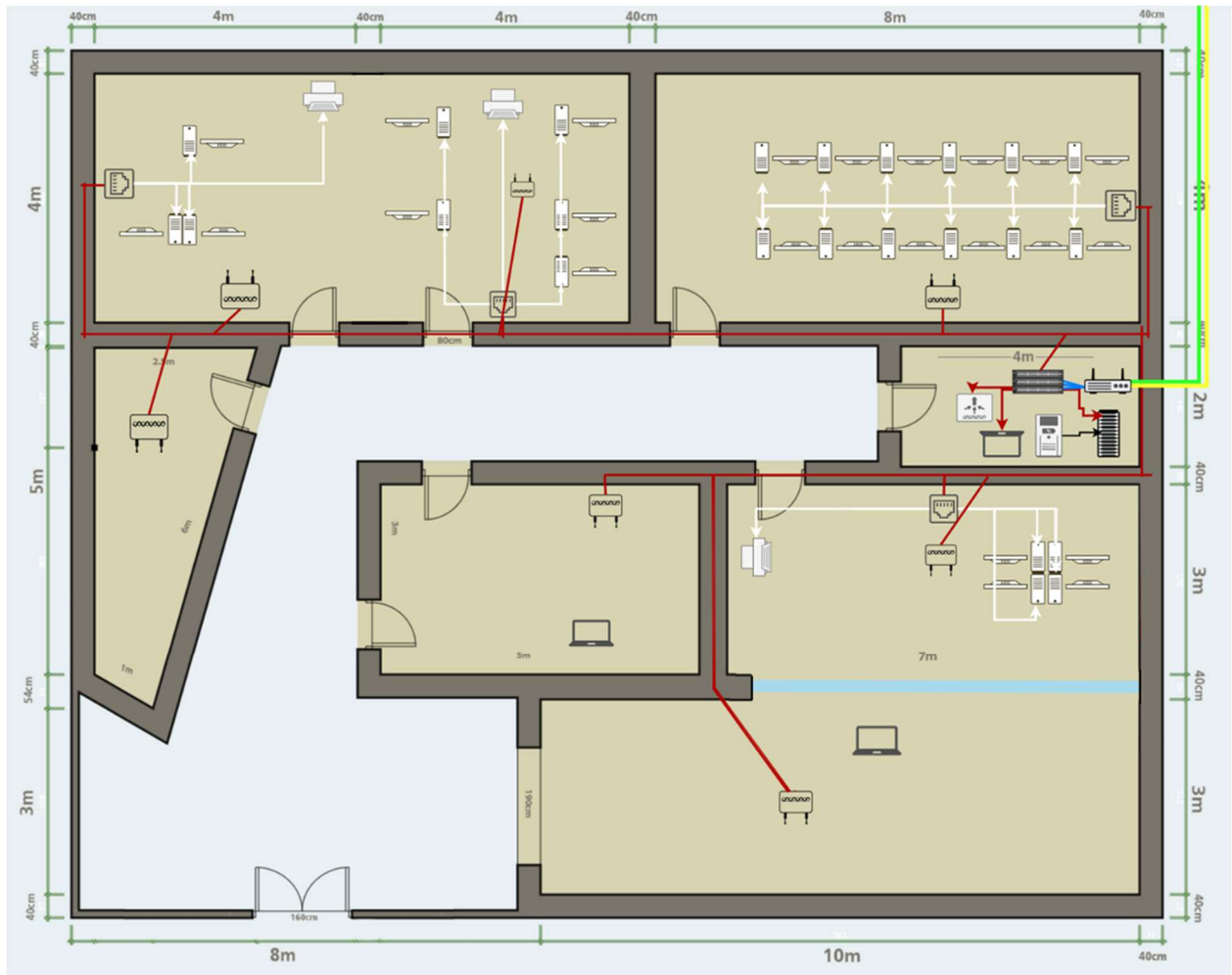


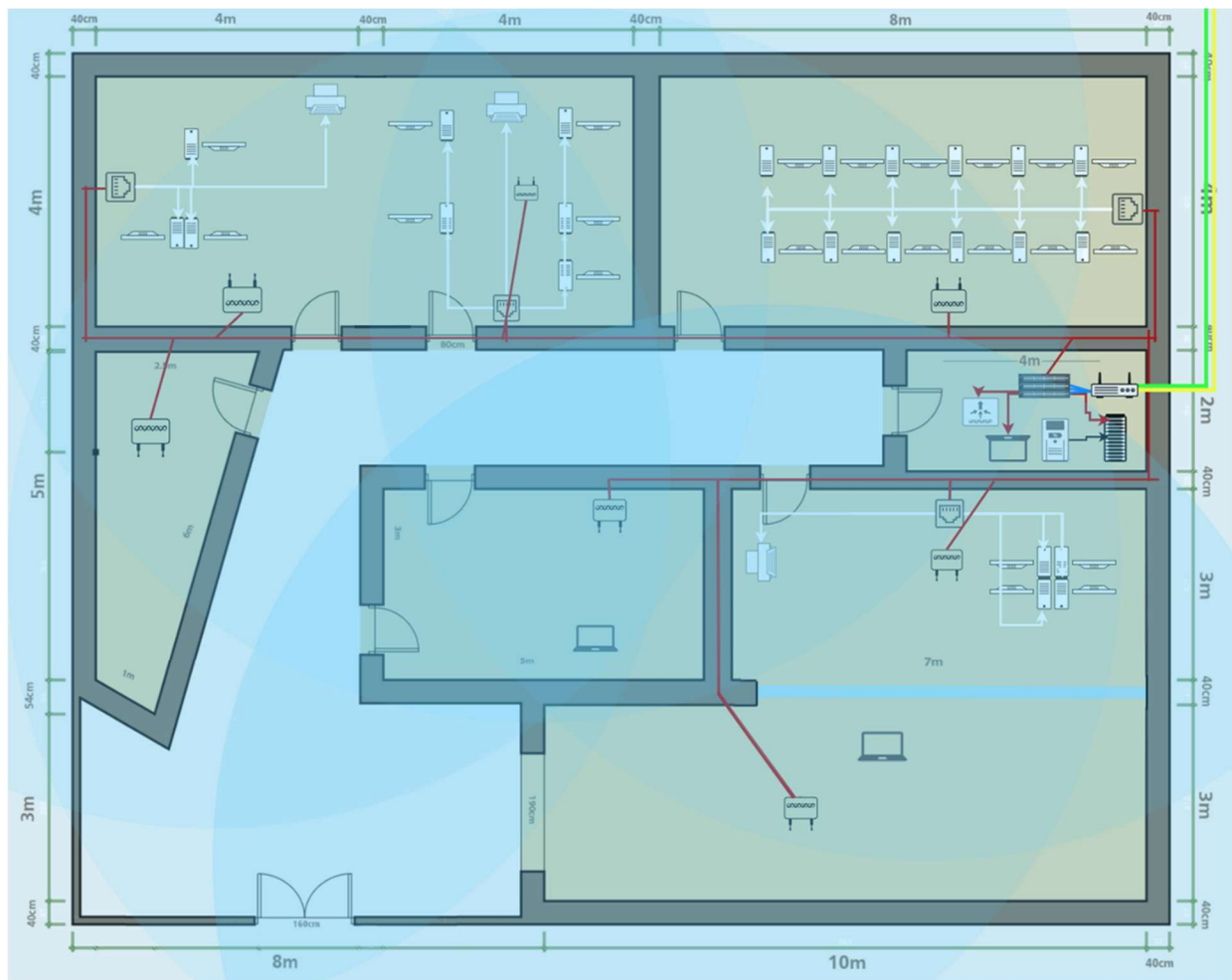
Fig1.1.1, Floor Plan Generated From Draw.io



Fig1.2, Floor Plan Generated From Draw.io

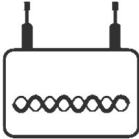


Fig(1.3), Network Design Generated From Draw.io



Fig(1.4), Network Design Generated From Draw.io

Legend

Icon	Explanation
	Ethernet Copper (Cat6)
	Power Cables
	Single-Mode Fiber Optic ISP Cable
	DCE Serial Cable
	Ethernet cables (Cat6) With rj-45 connector
	Multi-Mode Fiber Optic Cable
	Glass Wall
	Computer
	Ethernet Switch Port
	Cisco Aironet 2802i
	HP Printers
	Cisco 2504 Wireless LAN Controller

Icon	Explanation
	Cisco Catalyst 2960X-48FPS-L Switches with 24 ports each
	APC Ups for the server
	Cisco UCS File Server
	Cisco 2911 Router
	Laptop

All icons in this legend are sourced from draw.io

Justification

The branch occupies a nearly square-shaped floor plan with a total area of approximately **120–130 m²**, which makes it suitable for a medium-sized working environment. The layout includes **3 main departments**.

The **Administration Office** covers around **32 m² (4 m × 8 m)**, providing enough space for the manager's desk, visitor seating, and private discussions, the department's jobs are handling employee management, recruitment, and welfare. The **Finance Department** is slightly smaller because unlike the previous department the finance department focus on workstations and storage, at about 21m² (7 m × 3 m) accommodating 4 employees along with filing cabinets and financial records. The **IT Department**, which also supports server equipment, occupies **32m² (8 m × 4 m)** and hosts **12 members**, ensuring technical operations, data handling, and system support with the benefit of being directly connected with the server room which provides the whole floor with network.

A **Meeting Room** of about **15 m² (5 m × 3 m)** is included, designed to seat 8–10 people comfortably for presentations and discussions. The **Reception Area** at the entrance, also around **30 m² (3 m × 10 m)**, acts as the welcoming space for visitors, with a reception desk and waiting chairs. To support employee well-being, a **Resting Area** of approximately **9–10 m²** (irregular triangular layout) is available, furnished with sofas and a small coffee table.

LAN/WAN Topology Justification

We will use a **star topology** for the LAN and **point-to-point** WAN links.

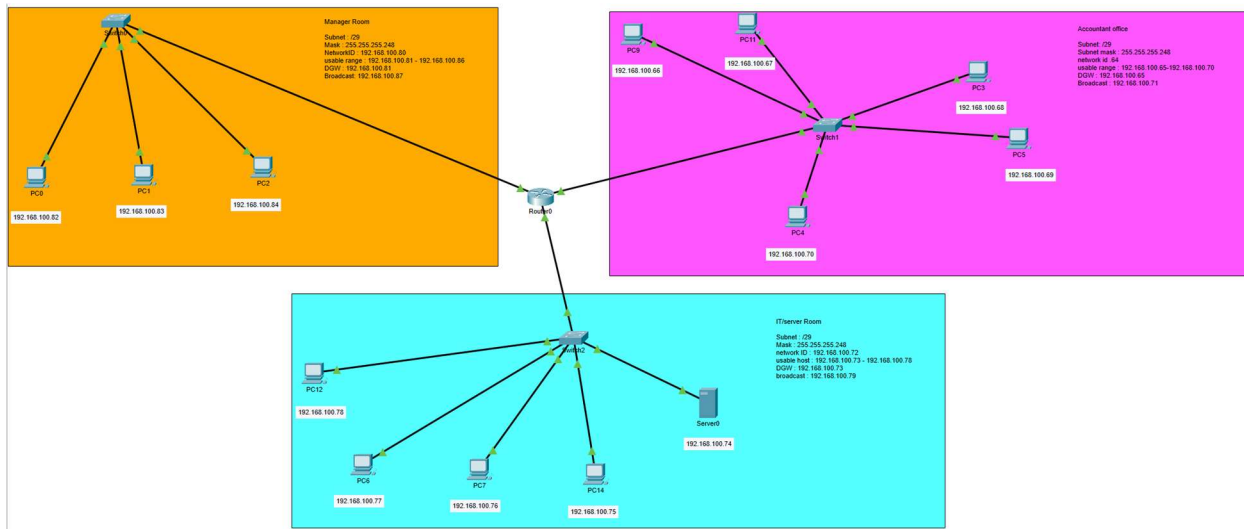


Fig 6.1, Johor Bahru Branch LAN, Example Generated From Cisco Packet Tracer 8.2.2

Star Topology uses access switches in the main areas and uplinks to the distribution/core layer, meaning that all faults are **limited** to one spur connection, moves/adds/changes are easy, and all connections can implement the same security and **QoS** measures at a consistently designed point of aggregation. This also complies with structured cabling and ensures expected performance for **full-duplex** Ethernet capabilities.

VLSM Box

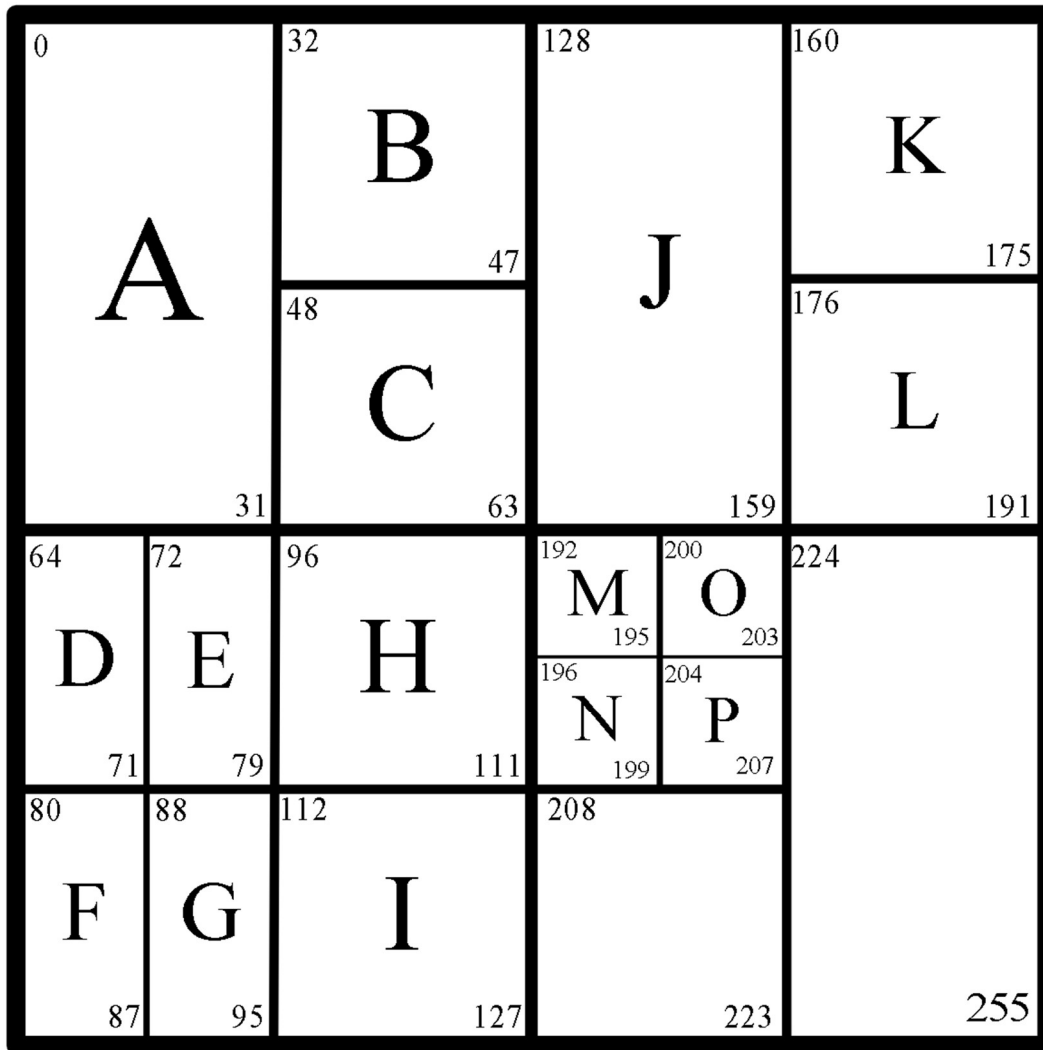


Fig7.0, VLSM Box Generated From Microsoft Visio

Blocks Defining

Block	Branch	Area / Department
A	Kuala Lumpur	Server Room/ IT Department
B	Kuala Lumpur	Management Office/ Resting Area
C	Kuala Lumpur	Finance Department
D	Johor Bahru	Server Room/ IT Department
E	Johor Bahru	Accountant Office
F	Johor Bahru	Management Office
G	Pulau Pinang	Office Room
H	Pulau Pinang	IT Room
I	Pulau Pinang	Management Room
J	Kota Kinabalu	IT Department
K	Kota Kinabalu	HR & Management Office
L	Kota Kinabalu	Server Room
M	Kuala Lumpur	Router
N	Johor Bahru	Router
O	Pulau Pinang	Router
P	Kota Kinabalu	Router

IP Addressing Plan

Network IP: 192.168.100.0

IP Addressing for all mentioned branches.

Block	Subnet	Subnet Mask	Network ID	Usable Host Range	DGW	Broadcast
A	/27	.224	.0	.2 – 30	.1	.31
B	/28	.240	.32	.34 – .46	.33	.47
C	/28	.240	.48	.50 – .62	.49	.63
D	/29	.248	.64	.66 – .70	.65	.71
E	/29	.248	.72	.74 – .78	.73	.79
F	/29	.248	.80	.82 – .86	.81	.87
G	/29	.248	.88	.90 – .94	.89	.95
H	/28	.240	.96	.98 – .110	.97	.111
I	/28	.240	.112	.114 – .126	.113	.127
J	/28	.240	.128	.130 – .158	.129	.159
K	/29	.248	.160	.162 – .174	.161	.175
L	/29	.248	.176	.178 – .190	.177	.191
M	/30	.252	.192	.193/.194	N/A	.195
N	/30	.252	.196	.197/.198	N/A	.199
O	/30	.252	.200	.201/.202	N/A	.203
P	/30	.252	.204	.205/.206	N/A	.207

Kuala Lumpur Branch Router

Network IP: 192.168.100.0

Device	Interface	Network Address	IP Address	Usable Address Range	Broadcast Address	Default Gateway
KL Router	Gig0/0	.0/27	.97	.2 - .30	.31	Not Applicable
	Gig0/2	.32/28	.17	.34 - .46	.47	Not Applicable
	Gig0/1	.48/28	.33	.50 - .62	.63	Not Applicable
	S0/0/0	.192/30	.193	-	.195	Not Applicable
	S0/0/1	.200/30	.202	-	.203	Not Applicable

References

APDP. (2010). *Principles of Personal Data Protection*. Retrieved from PERSONAL DATA PROTECTION OFFICE: <https://www.pdp.gov.my/ppdpv1/en/principles-of-personal-data-protection>

Campus LAN and Wireless LAN Solution Design Guide. (4 May, 2020). Retrieved from Cisco: <https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html>

Campus LAN and Wireless LAN Solution Design Guide. (2020). Retrieved from Cisco: <https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html>

IP Routing Configuration Guide, Cisco IOS XE 17.x. (2 Nov, 2022). Retrieved from Cisco: https://www.cisco.com/c/en/us/td/docs/routers/ios/config/17-x/ip-routing/b-ip-routing/m_iri-iprouting.html

ISO/IEC 27000 family Information security management. (2022). Retrieved from ISO: <https://www.iso.org/standard/iso-iec-27000-family>

Khan, T. (n.d.). *What is network topology?* Retrieved from IBM: <https://www.ibm.com/think/topics/network-topology?>

LAN Topologies. (17 July, 2025). Retrieved from Hewlett Packard Enterprise: <https://arubanetworking.hpe.com/techdocs/VSG/docs/010-campus-design/esp-campus-design-041-lan-design-topologies>

Layer 2 Configuration Guide, Cisco IOS XE 17.15.x (Catalyst 9300 Switches). (14 Aug, 2024). Retrieved from Cisco: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9300/software/release/17-15/configuration_guide/lyr2/b_1715_lyr2_9300_cg/configuring_etherchannels.html

Murugiah Souppaya, K. S. (2022). *Guide to Enterprise Patch Management Planning: Preventive Maintenance for Technology*. Retrieved from NIST:
<https://csrc.nist.gov/pubs/sp/800/40/r4/final>

What is Cisco Packet Tracer? (n.d.). Retrieved from Cisco Networking Academy:
https://www.netacad.com/cisco-packet-tracer?utm_source=chatgpt.com